

The Lottery Ticket Hypothesis

Finding Sparse, Trainable Neural Networks

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A Starting Point

Language Models are Few-Shot Learners

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Abstract

Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

The Problem with Big Neural Networks

- Modern neural networks are becoming gigantic (take GPT-3 for example: 175B parameters, 700GB of weights, 4.6 million dollars to train!).
- With bigger networks, comes slower training and inference
- Modern neural networks are **massively over-parameterised**: far more parameters than strictly needed to fit the training data.
- **Post-hoc pruning** routinely removes **90%+** of weights *after* training, with little to no loss in accuracy (LeCun et al., 1990; Han et al., 2015).

Optimal Brain Damage

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ABSTRACT

We have used information-theoretic ideas to derive a class of practical and nearly optimal schemes for adapting the size of a neural network. By removing unimportant weights from a network, several improvements can be expected: better generalization, fewer training examples required, and improved speed of learning and/or classification. The basic idea is to use second-derivative information to make a tradeoff between network complexity and training set error. Experiments confirm the usefulness of the methods on a real-world application.

How Pruning Works

- 1. Train a dense network to convergence.
- 2. Prune a fraction of the weights (based on magnitude, sensitivity, etc.) to create a sparse subnetwork.
- 3. Fine-tune the sparse subnetwork to recover accuracy.
- 4. Repeat steps 2–3 iteratively to reach extreme sparsity.

Observation 1: the *final sparse subnetwork*, when trained in isolation, can reach the same accuracy as the original dense network.

Observation 2: *inference problem* is solved; the sparse network is smaller and faster to run.

But: The training bottleneck remains – we still need to train the full dense network first, which is expensive.

Research Question

If a small subnetwork is sufficient, can we just train that directly?

PRUNING FILTERS FOR EFFICIENT CONVNETS

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ABSTRACT

The success of CNNs in various applications is accompanied by a significant increase in the computation and parameter storage costs. Recent efforts toward reducing these overheads involve pruning and compressing the weights of various layers without hurting original accuracy. However, magnitude-based pruning of weights reduces a significant number of parameters from the fully connected layers and may not adequately reduce the computation costs in the convolutional layers due to irregular sparsity in the pruned networks. We present an acceleration method for CNNs, where we prune filters from CNNs that are identified as having a small effect on the output accuracy. By removing whole filters in the network together with their connecting feature maps, the computation costs are reduced significantly. In contrast to pruning weights, this approach does not result in sparse connectivity patterns. Hence, it does not need the support of sparse convolution libraries and can work with existing efficient BLAS libraries for dense matrix multiplications. We show that even simple filter pruning techniques can reduce inference costs for VGG-16 by up to 34% and ResNet-110 by up to 38% on CIFAR10 while regaining close to the original accuracy by retraining the networks.

The Core Question

Frankle & Carbin (ICLR 2019)

Does a randomly-initialised, dense network contain a **sparse subnetwork** that, trained **in isolation**, matches the full network's accuracy?

- If yes: over-parameterisation may exist to provide a **winning combination of initial weights**
- This would explain *why* pruning works, and could guide more efficient training.

THE LOTTERY TICKET HYPOTHESIS: FINDING SPARSE, TRAINABLE NEURAL NETWORKS

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ABSTRACT

Neural network pruning techniques can reduce the parameter counts of trained networks by over 90%, decreasing storage requirements and improving computational performance of inference without compromising accuracy. However, contemporary experience is that the sparse architectures produced by pruning are difficult to train from the start, which would similarly improve training performance.

We find that a standard pruning technique naturally uncovers subnetworks whose initializations made them capable of training effectively. Based on these results, we articulate the *lottery ticket hypothesis*: dense, randomly-initialized, feed-forward networks contain subnetworks (*winning tickets*) that—when trained in isolation—reach test accuracy comparable to the original network in a similar number of iterations. The winning tickets we find have won the initialization lottery: their connections have initial weights that make training particularly effective.

We present an algorithm to identify winning tickets and a series of experiments that support the lottery ticket hypothesis and the importance of these fortuitous initializations. We consistently find winning tickets that are less than 10-20% of the size of several fully-connected and convolutional feed-forward architectures for MNIST and CIFAR10. Above this size, the winning tickets that we find learn faster than the original network and reach higher test accuracy.

The Hypothesis – Formal Statement

Lottery Ticket Hypothesis (LTH)

A randomly-initialised, dense neural network $f(x; \theta_0)$ contains a subnetwork $f(x; m \odot \theta_0)$ such that, when trained in isolation, it can match the test accuracy of the original network after training for at most j iterations.

- The mask m is obtained by **magnitude pruning** *after* training.
- The crucial twist: surviving weights are **reset to θ_0** – not kept at their trained values.
- Such a pair (m, θ_0) is called a **winning ticket**.

Why “Lottery Ticket” ?

- Training a large, over-parameterised network is like **buying many lottery tickets** at once – many random sub-initialisations, in parallel, inside one big network.
- Pruning after training reveals which combination of weights was already “**lucky**” – the ticket that would have won on its own.
- Initialisation plays the role of the **lottery draw**: only certain combinations of initial weights are trainable once isolated.

Important nuance

Up until now, this is a statement about *existence*, not about how to find such tickets cheaply – that turns out to be the hard part (but we’ll discuss it!).

Finding A Winning Ticket – Iterative Magnitude Pruning (IMP)

Randomly initialise network: θ_0 , mask $m \leftarrow \mathbf{1}$;

for $round = 1, \dots, R$ **do**

 Train $f(x; m \odot \theta_0)$ for j iterations $\rightarrow$ trained weights θ_j ;

 Prune $p\%$ of the **smallest-magnitude** surviving weights in $\theta_j \rightarrow$ update m ;

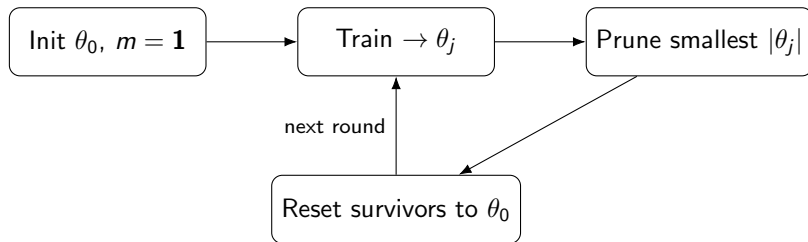
Reset surviving weights to θ_0 (not θ_j);

end

return (m, θ_0)

- One-shot pruning (prune to target sparsity in a single step) also works, but **iterative** pruning finds smaller winning tickets.
- Typical per-round pruning rate: $p \approx 20\%$ of remaining weights, layer-wise.

IMP as a Loop



Repeated until the target sparsity is reached
Each round yields a smaller candidate ticket

Experimental Setup

- **Networks:**
 - LeNet-300-100 (fully-connected) on MNIST.
 - Conv-2, Conv-4, Conv-6 (small VGG-style CNNs) on CIFAR-10.
- **Pruning:** Iterative Magnitude Pruning, layer-wise, $\sim 20\%$ per round.
- **Baselines:**
 - **Randomly reinitialised** sparse network, same mask m , fresh weights $\theta'_0 \neq \theta_0$.
 - Dense network trained normally.
- **Metric:** test accuracy vs. iterations, at each sparsity level.

Why Does Resetting to Initialisation Matter?

- Pruning + *fine-tuning* (no reset) gives a good *post-hoc* compressed model – but it is tied to the already-trained state.
- Resetting to θ_0 tests something stronger: that the **mask itself** identifies a trainable subnetwork *a priori*, before any of its own training.
- The mask acts like a **blueprint** already encoded in the initial random weights.

Key insight

Initialisation provides a **weight subspace** that is particularly amenable to optimisation; the mask selects that subspace.

A Motivating Picture



The Instability of Winning Tickets

- Tickets found at one setting do **not** necessarily remain winning tickets if we:
 - change the optimiser (e.g., SGD $\rightarrow$ Adam),
 - change the dataset,
 - change the architecture, even a width multiplier.
- They are **fragile**: tied jointly to initialisation, architecture, and training recipe.
- For networks beyond a certain depth/scale, standard (reset-to-init) IMP simply **fails to find** winning tickets at all – accuracy collapses.

This instability motivates a more robust variant of the hypothesis.

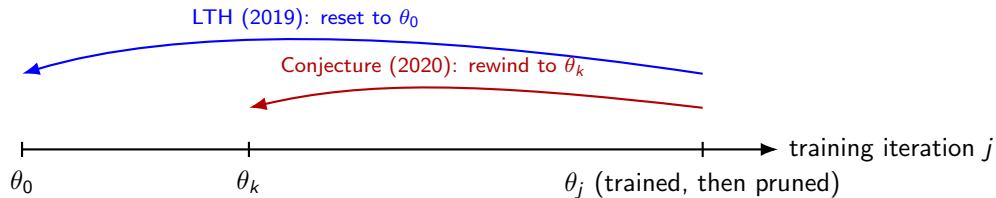
From Hypothesis to Conjecture: Rewinding

Lottery Ticket Conjecture (Frankle et al., 2020)

For large-scale networks (e.g., ResNet-50 on ImageNet), winning tickets can be found if surviving weights are **rewound** to an **early point in training** θ_k ($k \ll j$), rather than to θ_0 .

- Rewinding stabilises the mask and makes IMP feasible for modern architectures.
- Suggests the **early phase of training** establishes a “skeleton” of important connections, and some early SGD noise is needed to escape a bad initial basin.

Reset vs. Rewind



Rewinding to a small k (a few hundred/thousand iterations) is often enough to stabilise large-scale IMP.

What We Have Seen So Far

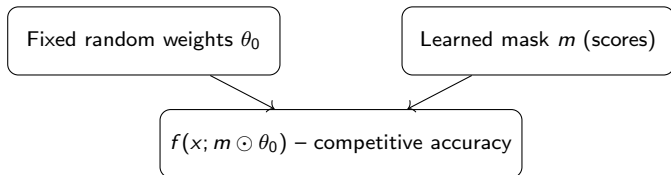
- Dense, randomly-initialised networks contain **sparse subnetworks** that can train to full accuracy in isolation.
- Such **winning tickets** are uncovered by iterative magnitude pruning combined with resetting/rewinding weights.
- The phenomenon highlights the **importance of initialisation**, not just architecture capacity.
- The **Lottery Ticket Conjecture** (rewinding) extends the idea to large-scale settings where naive reset fails.

What We'll Discuss Now

- Some Theoretical perspectives on why winning tickets exist and why rewinding helps.
- Practical directions for finding winning tickets cheaply, and for transferring them across tasks.
- Open challenges and limitations of the current understanding.

Supermask: Weight-Agnostic Winning Tickets

- Zhou et al. (2019, “Deconstructing Lottery Tickets”): even **without any weight training**, a well-chosen binary mask over *fixed random* weights can achieve surprisingly good accuracy.
- This **supermask** phenomenon isolates the contribution of the mask structure alone, separate from weight learning.



Linear Mode Connectivity

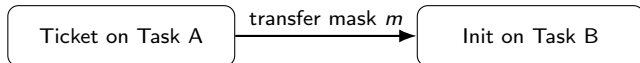
- Two networks are **linearly mode connected** if the linear interpolation of their trained weights has no loss barrier: $\mathcal{L}(\alpha\theta_A + (1 - \alpha)\theta_B)$ stays low for all $\alpha \in [0, 1]$.
- Winning tickets (found via rewinding) are often linearly connected to the **full trained network** – they sit in the same loss basin.
- This gives a **mechanistic explanation**: rewinding finds the point at which SGD noise has already committed the network to a single basin.

Can We Find Winning Tickets Early?

- **Combinatorial optimisation:** finding an *optimal* sparse mask is NP-hard in general; IMP is a greedy heuristic that works surprisingly well in practice.
- Standard IMP needs **multiple full train-prune cycles** – expensive, arguably defeating the purpose of finding an efficient network.
- “Pruning at initialisation” methods try to find a mask **before any training**:
 - **SNIP** (Lee et al., 2019): prune by connection sensitivity (gradient-based saliency), one-shot.
 - **GraSP** (Wang et al., 2020): preserve gradient flow through the pruned network.
 - **SynFlow** (Tanaka et al., 2020): data-free, avoids **layer collapse** at extreme sparsity.
- These methods are far cheaper, but typically **underperform** IMP at matched sparsity – an open gap.

Transfer and Universality

- Tickets found on one dataset sometimes **transfer** to related tasks and even across optimisers.
- Morcos et al. (2019): tickets found on large datasets (e.g., ImageNet) can transfer to and outperform tickets found directly on smaller datasets (e.g., CIFAR-10).
- Raises the prospect of a **universal** sparse mask, reusable across tasks – though this is far from settled.



Known Limitations and Critiques

- Results are **sensitive to hyperparameters** (learning rate, pruning rate, warm-up); several follow-up studies report difficulty reproducing the strongest claims at scale.
- IMP is **expensive**: it needs R full training runs to reach a target sparsity, which is worse than a single dense training run.
- Reset-to-init (the *original* 2019 claim) largely **fails beyond small/medium networks**; the practically useful version is rewinding, which is a materially weaker and later claim.
- Debate continues over whether the effect is really about **specific weight values** or about the **sparse architecture / signs** being found (cf. supermasks).

Open Challenges

1. Why does resetting/rewinding work, while random reinitialisation with the *same* mask fails?
2. Can winning tickets be found with (close to) zero training cost?
3. How does LTH interact with width/depth scaling laws and modern large-model training (e.g., Transformers)?
4. Can LTH-style analysis inform *designing* efficient architectures from scratch, rather than discovering them post hoc?

Where Is the Lottery Ticket Hypothesis Today?

- The original LTH result remains compelling, but its **strongest form does not scale cleanly** to modern networks.
- The practical version has shifted from **resetting to θ_0** toward **rewinding, early pruning, and sparse training**.
- A major bottleneck remains:
finding a winning ticket can cost almost as much as training the dense model itself.
- Recent work therefore asks a more fundamental question:

The Modern Sparse-Training Question

Can we find a good sparse architecture *before* expensive dense training, and train it successfully from a new random initialization?

- This reframes LTH from a hypothesis about **lucky initialisations** into a broader question about **which sparse structures make optimisation easy**.

Recent Directions: From Winning Tickets to Sparse Training

- **Sparse training from new initialisations**
 - Adnan et al. (ICML 2025) show that LTH masks are strongly coupled to the original optimisation basin.
 - By **aligning the mask using neural-network weight symmetries**, the same sparse structure can work much better with a new random initialization.
- **Transformers and language models**
 - LTH-style ideas are increasingly being studied beyond CNNs, including Transformer architectures and LLMs.
 - **KS-Lottery (NAACL 2025)** investigates certified lottery tickets for multilingual transfer in large language models.
- **Beyond vision: theory and new architectures**
 - ICML 2025 work on GNNs connects winning tickets to **expressivity**, providing theoretical conditions for sparse subnetworks to preserve predictive power.
 - Recent work also explores structured tickets, video representations, and more hardware-friendly forms of sparsity.

Take-Home Messages

- Dense, randomly-initialised networks contain **sparse subnetworks** that can train to full accuracy in isolation.
- Such **winning tickets** are uncovered by iterative magnitude pruning combined with resetting/rewinding weights.
- The phenomenon highlights the **importance of initialisation**, not just architecture capacity.
- The **Lottery Ticket Conjecture** (rewinding) extends the idea to large-scale settings where naive reset fails.
- Open problems remain around **cheaply finding** tickets and fully explaining **why** the effect occurs.

Thank you! Questions?

References I

-  J. Frankle and M. Carbin. *The Lottery Ticket Hypothesis: Finding Sparse, Trainable Neural Networks*. ICLR, 2019.
-  J. Frankle, G. K. Dziugaite, D. M. Roy, M. Carbin. *Linear Mode Connectivity and the Lottery Ticket Hypothesis*. ICML, 2020.
-  J. Frankle et al. *The Lottery Ticket Hypothesis at Scale*. arXiv:1903.01611, 2019.
-  H. Zhou, J. Lan, R. Liu, J. Yosinski. *Deconstructing Lottery Tickets: Zeros, Signs, and the Supermask*. NeurIPS, 2019.
-  A. Morcos, H. Yu, M. Paganini, Y. Tian. *One Ticket to Win Them All: Generalizing Lottery Ticket initialisations across Datasets and Optimizers*. NeurIPS, 2019.
-  N. Lee, T. Ajanthan, P. H. S. Torr. *SNIP: Single-Shot Network Pruning based on Connection Sensitivity*. ICLR, 2019.
-  C. Wang, G. Zhang, R. Grosse. *Picking Winning Tickets Before Training by Preserving Gradient Flow*. ICLR, 2020.

References II

-  H. Tanaka, D. Kunin, D. L. K. Yamins, S. Ganguli. *Pruning Neural Networks without Any Data by Iteratively Conserving Synaptic Flow*. NeurIPS, 2020.
-  Y. LeCun, J. Denker, S. Solla. *Optimal Brain Damage*. NeurIPS, 1990.
-  S. Han, J. Pool, J. Tran, W. Dally. *Learning both Weights and Connections for Efficient Neural Networks*. NeurIPS, 2015.
-  M. Adnan, R. Jain, E. Sharma, R. G. Krishnan, Y. Ioannou. *Sparse Training from Random Initialization: Aligning Lottery Ticket Masks using Weight Symmetry*. ICML, 2025.
-  F. Yuan, C. Ma, S. Yuan, Q. Sun, L. Li. *KS-Lottery: Finding Certified Lottery Tickets for Multilingual Transfer in Large Language Models*. NAACL-HLT, 2025.
-  L. Kummer, S. Moustafa, A. Ehrlich, F. Bause, N. Suess, W. N. Gansterer, N. M. Kriege. *Weisfeiler and Leman Go Gambling: Why Expressive Lottery Tickets Win*. ICML, 2025.
-  G. Vasudev, A. Adithya. *The EarlyBird Gets the WORM: Heuristically Accelerating EarlyBird Convergence*. NeurIPS Efficient Natural Language and Speech Processing Workshop, 2024.

References III

-  T. J. Saleem, R. Ahuja, S. Prasad, B. Lall. *Insights into the Lottery Ticket Hypothesis and Iterative Magnitude Pruning*. arXiv:2403.15022, 2024.